A Quartic Period Torsor and Optimal Positive Moment Bounds

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Abstract

The stable quartic integral has a rank-two differential period torsor. We describe its local realizations, their combinatorial expansions, and the additional data needed to recover the real integral. Every finite Hermitian matrix size is a matrix coefficient of a symmetric power of the same system. To compare approximations, we classify positive generalized Stieltjes representations: a positive homogeneous interaction of degree $2m$ in d Gaussian variables has exact order $d/(2m)$. The smallest admissible order gives the strongest Gauss and Gauss–Radau bounds from a fixed number of perturbative coefficients. For quartic and sextic interactions these bounds converge at every positive coupling. In particular, they give certified approximations to all scalar sextic correlation functions. At degree eight and above, a moment-indeterminacy example shows why positivity and all perturbative coefficients can still be insufficient. Exact computations illustrate the bounds for Hermitian matrices of sizes one through four. Worked examples on finite rewrite graphs and hypergraph incidence graphs express nonlinear field correlations through the same quartic periods and spanning-forest counts.

1 Introduction

An integral may have several combinatorial expansions with very different numerical behavior. A useful common object should explain their relations, specify the information each expansion retains, and help decide which one to use. A torsor of period comparisons suggests such an object, but the integral value itself is not a torsor. Even in a simple example one must distinguish the differential equation, a basis of solutions, an integration cycle, and an approximation to the chosen period.

We work first with

$$q_j(t) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} x^j \exp\left(-\frac{x^2}{2} - \frac{tx^4}{4}\right) \mathrm{d}x, \qquad t > 0. \tag{1.1}$$

The two moments q_0, q_2 satisfy a rational differential system. Its horizontal frames form a differential torsor. The Gaussian expansion at $t = 0$, the quartic expansion at infinity, and Taylor expansions at ordinary points are local realizations of this system. The formal local classification is complete in this specified category; it is not a classification of arbitrary differential extension fields.

There is also a second, useful classification. Suppose that the Wick coefficients are used to reconstruct a positive integral of the form $\int (1 + \lambda s)^{-\beta} \, \mathrm{d}\rho(s)$. Which orders β are possible, and which gives the best bounds? For the normalized Hermitian quartic partition function of size M,

the answer is $\beta \geq M^2/4$. The minimal order gives stronger moment bounds than every larger order, at the same number of input coefficients and at every positive coupling. Here "best" is an extremal statement within a stated class of positive representations, for fixed input moments.

The ingredients have substantial precedents. Picard–Vessiot torsors and formal local differential equations are standard [20]. Comparison torsors for periods are treated by Brown [4]; exponential integrals belong to the framework of Fresán and Jossen [8]. Wick expansions and moment determinants are standard matrix-model tools [6], and the quartic weight has a well-developed orthogonal-polynomial and Painlevé theory [5]. Generalized positive-kernel approximants already appear in Savit's work on lattice theories [19, Appendix B]. Generalized Stieltjes classes and their exact orders are studied by Sokal and by Karp–Prilepkina [21, 11]; positive quadrature bounds are classical [9, 15]. We give elementary proofs of the specific representation and order comparison used here.

Fiol, Gijón and Lejarza Alonso recently proved convergence of rational Schwinger–Dyson approximants for all scalar quartic correlation functions [7]. Their introduction asks about other even potentials and about products of Padé approximants. Our sextic corollary addresses convergence by a different construction: positive quadrature of reweighted Wick moments. It does not identify the truncations of the sextic Schwinger–Dyson hierarchy or prove that they are products of Padé approximants.

The graph-theoretic motivation comes from the local Whitehead chain and its stationary measure, weighted by inverse automorphism orders [13]. The present paper keeps that motivation separate from the analysis of the integral. In particular, a restriction to some graphs is not automatically a specialization of a period torsor, nor does it automatically bound the error in an integral. Section 8 gives a concrete application to finite graphs of rewrite histories and to hypergraph incidence graphs.

2 The differential torsor

For a rapid-decay contour γ, put $m_j^\gamma = \int_\gamma x^j e^{-x^2/2-tx^4/4}\,\mathrm{d}x$. Integration by parts and differentiation give

$$tm_{j+4} = (j+1)m_j - m_{j+2}, \qquad (m_j)' = -\tfrac{1}{4}m_{j+4}. \tag{2.1}$$

Thus the even moments form a rank-two system

$$v' = A(t)v, \qquad A(t) = \begin{pmatrix} -1/(4t) & 1/(4t) \\ 1/(4t^2) & -(1+3t)/(4t^2) \end{pmatrix}, \qquad v = \begin{pmatrix} m_0 \\ m_2 \end{pmatrix}. \tag{2.2}$$

Equivalently,

$$16t^2y'' + (32t+4)y' + 3y = 0, \qquad m_2 = 4ty' + y. \tag{2.3}$$

The full quartic twisted de Rham system also has an odd-moment part. Throughout, the torsor refers to the even-moment subsystem (2.2), generated by $\mathrm{d}x, x^2\,\mathrm{d}x$.

Let $K = \overline{\mathbb{Q}}(t)$, with derivation $D = \mathrm{d}/\mathrm{d}t$. The coordinate algebra

$$\mathcal{R} = K[y_{11}, y_{12}, y_{21}, y_{22}, \det(Y)^{-1}], \qquad DY = AY, \tag{2.4}$$

is the frame torsor with its differential structure. Explicitly, for a differential extension F/K, its horizontal points are

$$\mathcal{T}^D(F) = \{Y \in \mathrm{GL}_2(F) : DY = AY\}.$$

If this set has a point, it has a free transitive right action of $\mathrm{GL}_2(F^D)$. Indeed, $D(Y_1^{-1}Y_2) = 0$ for any two horizontal frames. This is the torsor property: two frames differ by one uniquely determined constant matrix. The underlying algebraic GL_2-torsor is trivial. The differential structure and the selected period point carry the information.

Proposition 2.1. *The differential Galois group of* (2.2) *over* K *is* GL_2. *In particular,* (2.4) *is its Picard–Vessiot torsor algebra.*

Proof. Set $z = 1/(8t)$ and $y = t^{-1/2}e^z f(z)$. Equation (2.3) becomes the modified Bessel equation of order $1/4$. The Bessel group at this order is SL_2 [20, Example 1.3.32]; replacing its variable by iz gives the modified equation. The algebraic gauge and exponential scalar twist do not change its projective group, which is consequently PGL_2. On the other hand,

$$D \log \det Y = \operatorname{Tr} A = -\frac{1}{t} - \frac{1}{4t^2}, \qquad \det Y = c\, t^{-1} e^{1/(4t)}. \tag{2.5}$$

This determinant has group $\mathbb{G}_m$: no nonzero power of the exponential is algebraic over K. The identity component of the full group has projective image PGL_2. A connected three-dimensional subgroup with that projective image would have finite scalar kernel, and hence no nontrivial character to $\mathbb{G}_m$. The determinant image rules this out. The group therefore has dimension four and equals GL_2. The Picard–Vessiot torsor theorem [20, Proposition 1.3.20] now identifies its algebra with (2.4). □

For $t > 0$, the contours $\mathbb{R}$ and $i\mathbb{R}$, oriented upwards, give a period frame

$$P = \begin{pmatrix} m_0^{\mathbb{R}} & m_0^{i\mathbb{R}} \\ m_2^{\mathbb{R}} & m_2^{i\mathbb{R}} \end{pmatrix}. \tag{2.6}$$

These two columns will be seen to be independent below. They are points over a complex analytic differential field; the gamma values in their comparison matrix require extension of the constant field. This differential torsor is not being identified with a full arithmetic motivic comparison torsor at a fixed value of t. That would require additional realization data and an arithmetic construction.

3 Classification of the local realizations

We allow completions at places of K, finite algebraic ramification, and formal changes of basis. We also record when an exponential extension is necessary. These operations give the following exhaustive list in this local category.

Theorem 3.1. *For* (2.2) *the local possibilities are as follows.*

1. *Every finite* $a \neq 0$ *is ordinary. There is a unique horizontal frame* $U_a = I + O(t-a)$ *over* $\overline{\mathbb{Q}}[[t-a]]$.
2. *At* $t = 0$, *the formal scalar solutions have exponential factors* 1 *and* $e^{1/(4t)}$. *A full frame exists over* $\overline{\mathbb{Q}}((t))(e^{1/(4t)})$, *but over no finite algebraic extension of* $\overline{\mathbb{Q}}((t))$.
3. *At* $t = \infty$, *the equation is regular singular with scalar exponents* $1/4, 3/4$ *in* $w = 1/t$. *A full frame exists over* $\overline{\mathbb{Q}}((w^{1/4}))$. *In a cover* $w = s^e$, *it exists over* $\overline{\mathbb{Q}}((s))$ *if and only if* $4 \mid e$.

The analytic problem at zero also needs sector and Stokes information.

Proof. The first assertion follows by substituting $U_a = \sum_{n \geq 0} U_n (t-a)^n$ in (2.2). If $A = \sum A_j (t-a)^j$, then

$$(n+1)U_{n+1} = \sum_{j=0}^{n} A_j U_{n-j}, \qquad U_0 = I. \tag{3.1}$$

At zero put

$$\widehat{q}(t)=\sum_{n\geq 0}(-t)^n a_n, \qquad a_n=\frac{(4n-1)!!}{4^n n!}=\frac{4^n(1/4)_n(3/4)_n}{n!}. \tag{3.2}$$

Direct substitution gives the two scalar solutions $\widehat{q}(t)$ and $e^{1/(4t)}\widehat{q}(-t)$. Applying $4tD+1$ to form their second components gives a frame of determinant $-t^{-1}e^{1/(4t)}$. Its exponential determinant cannot belong to any algebraic Laurent extension: its logarithmic derivative has a double pole, whereas that of an algebraic Puiseux series has at most a simple pole. This also proves the claimed obstruction.

At infinity, substituting $y=w^\rho(1+O(w))$ in (2.3) gives $(4\rho-1)(4\rho-3)=0$. The exponents differ by a noninteger, so there are two logarithm-free Frobenius series. Their fractional parts require ramification divisible by four. This proves necessity and sufficiency. The explicit frame below also proves sufficiency. These are instances of the usual formal local classification [20, Section 1.4]. □

The coefficients in (3.2) count pairings of $4n$ Gaussian half-edges, divided by the quartic vertex and exponential factors. They do not select all contours. For the real contour the series is asymptotic, and Taylor's integral remainder gives

$$\left|q_0(t)-\sum_{n=0}^{K}(-t)^n a_n\right|\leq t^{K+1}a_{K+1}. \tag{3.3}$$

Even partial sums are upper bounds, and odd partial sums are lower bounds. The terms eventually increase at every $t>0$.

At infinity, scaling $x=t^{-1/4}y$ gives the convergent series

$$\begin{aligned} q_0(t)&=\frac{t^{-1/4}}{2\sqrt{\pi}}\sum_{n\geq 0}\frac{(-t^{-1/2})^n}{n!}\Gamma(n/2+1/4),\\ q_2(t)&=\frac{t^{-3/4}}{\sqrt{\pi}}\sum_{n\geq 0}\frac{(-t^{-1/2})^n}{n!}\Gamma(n/2+3/4). \end{aligned} \tag{3.4}$$

The sums are entire in $t^{-1/2}$. Dominated convergence follows from $e^{-y^4/4+Cy^2}$ on each compact parameter set. Truncating either sum after degree K has absolute remainder at most its positive degree-$(K+1)$ term, including the displayed prefactor.

For completeness, let

$$A_0={}_1F_1(1/4;1/2;1/(4t)), \quad B_0={}_1F_1(3/4;3/2;1/(4t)),$$

and let C_0, D_0 have parameters $(5/4;3/2)$ and $(3/4;1/2)$, respectively, at the same argument. A rational-coefficient local frame is

$$Y_\infty=\begin{pmatrix} t^{-1/4}A_0 & t^{-3/4}B_0\\ -\frac{1}{2}t^{-5/4}C_0 & -2t^{-3/4}D_0\end{pmatrix}, \qquad \det Y_\infty=-2t^{-1}e^{1/(4t)}. \tag{3.5}$$

The actual period frame is

$$P=\frac{1}{\sqrt{2}}Y_\infty\begin{pmatrix}\Gamma(1/4) & i\Gamma(1/4)\\ -\Gamma(3/4) & i\Gamma(3/4)\end{pmatrix}. \tag{3.6}$$

In particular, $\det P=-2\sqrt{2}\pi i\,t^{-1}e^{1/(4t)}$. The rational local frame and the contour-dependent comparison constants have distinct roles.

The strong coefficients have a weighted combinatorial encoding, rather than the same Wick interpretation. For example, $4^n(1/4)_n = \prod_{j<n}(4j+1)$ counts histories of increasing plane 5-ary trees: insertion at one free leaf replaces it by five, increasing the number of available leaves by four. The product $\prod_{j<n}(4j+3)$ starts with three ordered leaves. The factorials and powers in ${}_1F_1$ supply the weights. Equation (3.1) likewise expands into finite weighted compositions of rational matrices. These descriptions show exactly how different combinatorial series can come from the same differential object.

4 All finite Hermitian matrix sizes

Let $\mathcal{H}_M$ be the real Euclidean space of Hermitian matrices, with inner product $\operatorname{Tr}(XY)$. Its dimension is $d = M^2$. Define

$$Z_M(\lambda) = \frac{\int_{\mathcal{H}_M} e^{-M\operatorname{Tr}(X^2)/2 - M\lambda\operatorname{Tr}(X^4)/4}\,\mathrm{d}X}{\int_{\mathcal{H}_M} e^{-M\operatorname{Tr}(X^2)/2}\,\mathrm{d}X}, \qquad \lambda > 0. \tag{4.1}$$

The eigenvalue determinant formula [6, Section 5.1], followed by $x \mapsto x/\sqrt{M}$, gives

$$Z_M(\lambda) = \frac{\det[q_{i+j}(\lambda/M)]_{0\le i,j<M}}{\prod_{j=0}^{M-1} j!}. \tag{4.2}$$

Here odd moments vanish. The denominator is the Gaussian moment determinant. By (2.1), every entry is a rational linear combination of q_0, q_2. Thus Z_M is a nonzero homogeneous polynomial of degree M in the same two periods, after the rational pullback $t = \lambda/M$.

Proposition 4.1. *The minimal homogeneous linear differential equation for Z_M over $\overline{\mathbb{Q}}(\lambda)$ has order exactly $M+1$. The associated module is Sym^M of the rank-two quartic module. Its formal exponential factors at zero are $e^{j/(4t)}$, $0 \le j \le M$.*

Proof. Homogeneous monomials of degree M in q_0, q_2 span a differential module of rank $M+1$. The two periods are algebraically independent over K by Proposition 2.1: after extending constants to $\mathbb{C}$ the group is still GL_2, and a first column of its generic frame has no polynomial relation. Thus the indicated monomials are linearly independent. The representation Sym^M of GL_2 is irreducible in characteristic zero. Consequently the differential submodule generated by the nonzero polynomial (4.2) is the whole module, so its cyclic equation has order $M+1$. The exponential factors follow by taking the symmetric power of the two formal factors at zero. □

At size two, for example, $Z_2(\lambda) = q_0(\lambda/2)q_2(\lambda/2)$, and elimination gives

$$8\lambda^4 Z_2''' + 4\lambda^2(16\lambda+3)Z_2'' + 2(55\lambda^2+18\lambda+2)Z_2' + 3(10\lambda+3)Z_2 = 0. \tag{4.3}$$

This has been checked symbolically against the symmetric-square system. The determinant and holonomic methods are consistent with, and do not replace, the established Painlevé description of quartic weights [5].

Write $Z_M \sim \sum_{n\ge0}(-\lambda)^n w_{M,n}$. With Gaussian covariance $\mathbb{E}[X_{ij}X_{kl}] = M^{-1}\delta_{il}\delta_{jk}$, Wick's rule yields

$$w_{M,n} = \frac{1}{4^n n!}\sum_{\alpha} M^{F(\sigma\alpha)-n}. \tag{4.4}$$

Here σ is a product of n specified 4-cycles, α runs over pairings of their $4n$ darts, and F counts permutation cycles. The sum includes disconnected diagrams. In particular, $w_{M,0} = 1$ and $w_{M,1} = (2M^2+1)/4$. Connected diagrams instead describe $\log Z_M$.

This matters when choosing restrictions. Retaining a genus range changes the coefficients in (4.4); it need not retain all moment positivity conditions used below. Likewise, Tanasa's multi-matrix schemes retain color, strand and insertion conditions [22, 3]. Such conditions must be imposed before a graph sum is identified with a particular integral. A formal interaction which is not positive cannot be assigned the positivity bounds of this paper.

5 Classification by positive kernel order

The next theorem gives a second precise meaning to the choice of a restriction. It concerns a class of positive coefficient realizations and resummations, not a literal base change of fields.

Definition 5.1. A normalized positive generalized Stieltjes representation of order $\beta > 0$ is

$$f(\lambda) = \int_{[0,\infty)} (1+\lambda s)^{-\beta} \, \mathrm{d}\rho_\beta(s), \qquad \rho_\beta([0,\infty)) = 1. \tag{5.1}$$

Its exact order is the infimum of admissible β.

This is the normalized convention for generalized Stieltjes functions [21, 11, 12]. If all moments exist, its unsigned asymptotic coefficients and representing moments satisfy

$$f(\lambda) \sim \sum_{n\geq 0} (-\lambda)^n w_n, \qquad \mu_{\beta,n} := \int s^n \, \mathrm{d}\rho_\beta(s) = \frac{n! w_n}{(\beta)_n}. \tag{5.2}$$

Thus a choice of order prescribes a particular reweighting of the same combinatorial coefficients.

Theorem 5.2. *Let Q be a strictly positive homogeneous polynomial of degree $2m$ on $\mathbb{R}^d \setminus \{0\}$, with $m \geq 2$, and let X be a standard Gaussian vector. For $f_Q(\lambda) = \mathbb{E}e^{-\lambda Q(X)}$, the admissible orders in (5.1) are exactly*

$$\beta \in [d/(2m), \infty).$$

Put $a = d/(2m)$. A representing random variable at the minimal order is

$$T = (2m)^m Q(\Omega) \prod_{j=1}^{m-1} G_{a+j/m}, \tag{5.3}$$

where Ω is uniform on the unit sphere and the G_b are independent gamma variables of shape b and scale one, independent of Ω. For every $\beta > a$, a representing variable is TB, with independent $B \sim \mathrm{Beta}(a, \beta - a)$.

Proof. Write $X = R\Omega$. Then $S = R^2/2$ has the gamma law of shape $d/2 = ma$, independently of Ω. Gamma multiplication gives the identity in law

$$S^m \overset{\text{law}}{=} m^m \prod_{j=0}^{m-1} G_{a+j/m}. \tag{5.4}$$

For rigor, this identity follows from the gamma multiplication formula at imaginary Mellin arguments, hence from equality of the characteristic functions of the logarithms. It does not assume that positive

integer moments determine the law. Since $Q(X) = 2^m S^m Q(\Omega)$, conditioning on the gamma variable G_a proves $f_Q = \mathbb{E}(1+\lambda T)^{-a}$.

Euler's beta identity gives

$$(1+\lambda s)^{-a} = \mathbb{E}_B(1+\lambda s B)^{-\beta}, \qquad \beta > a, \tag{5.5}$$

which proves inclusion in every larger order. This is the positive probability form of the usual inclusion of Stieltjes classes [11, Theorem 3].

Finally, scaling $X = \lambda^{-1/(2m)} Y$ gives

$$\lambda^a f_Q(\lambda) \longrightarrow c_Q = (2\pi)^{-d/2} \int_{\mathbb{R}^d} e^{-Q(Y)}\, \mathrm{d}Y \in (0,\infty). \tag{5.6}$$

Strict positivity on the sphere justifies dominated convergence and finiteness. If a representation of order $\beta < a$ existed, monotone convergence would give $\lambda^\beta f_Q(\lambda) \to \int s^{-\beta}\, \mathrm{d}\rho_\beta(s) > 0$, possibly infinite. This contradicts (5.6). □

Corollary 5.3. *The exact order for* (4.1) *is* $M^2/4$. *It is an ordinary Stieltjes function if and only if* $M = 1$ *or* $M = 2$. *Its minimal-order representing variable is*

$$T_M = \frac{4}{M} \operatorname{Tr}(\Omega^4) G_{(M^2+2)/4}, \qquad \operatorname{Tr}(\Omega^2) = 1. \tag{5.7}$$

Proof. Set $Y = \sqrt{M}X$ in (4.1) and apply the theorem to $Q(Y) = \operatorname{Tr}(Y^4)/(4M)$ on its M^2 real coordinates. On the unit sphere, $1/M \le \operatorname{Tr}(\Omega^4) \le 1$. □

Already this classification excludes a tempting approximation argument: at size three, ordinary Stieltjes–Padé convergence cannot be invoked for Z_3. The obstruction is its decay $\lambda^{-9/4}$, rather than a failure of complete monotonicity. The generalized positive representation remains available.

6 Choosing the strongest positive moment restriction

Fix an admissible order β and suppose that the coefficients $w_0, \dots, w_{2n}$ are known. From (5.2) construct the n-node Gauss rule for ρ_β, denoted $G_{\beta,n}$, and the $(n+1)$-node left Gauss–Radau rule $R_{\beta,n}$ with fixed node zero. The latter's nonzero nodes are the Gauss nodes for $s\, \mathrm{d}\rho_\beta(s)$. Apply both rules to $h_\beta(s) = (1+\lambda s)^{-\beta}$ and write $L_{\beta,n} = G_{\beta,n}(h_\beta)$ and $U_{\beta,n} = R_{\beta,n}(h_\beta)$. Assume the relevant Hankel moment matrices are positive definite; the measures in Theorem 5.2 have this property.

Lemma 6.1. *For every* $\lambda > 0$,

$$L_{\beta,n} \le f(\lambda) \le U_{\beta,n}.$$

Moreover, $L_{\beta,n}$ *is the smallest possible value of* $\int h_\beta\, \mathrm{d}\nu$ *over positive measures sharing moments through degree* $2n-1$, *and* $U_{\beta,n}$ *is the largest possible value over those sharing moments through degree* $2n$.

Proof. Hermite interpolation at the Gauss nodes, with multiplicity two, has degree at most $2n-1$. Its remainder has the sign of $h_\beta^{(2n)} > 0$ times the square of the nodal polynomial. It therefore lies below h_β on $[0,\infty)$. Integrating against any measure with the prescribed moments gives the lower bound. The discrete Gauss measure has those moments and attains equality, proving the extremal assertion.

For Radau interpolation use the simple node zero and double nonzero nodes. The remainder has the sign of $s h_\beta^{(2n+1)} < 0$ times a square, so the interpolant is an upper bound. The discrete

Radau measure matches moments through degree $2n$ and attains it. All integrals exist since the interpolants have the specified degrees. This is the classical sign mechanism for positive quadrature bounds [9, 15]. □

Theorem 6.2 (Order selection)**.** *Suppose that f has a representation of order $a > 0$ with all moments finite. For $a \le \beta_1 < \beta_2$, form the moments at each order from the same coefficients w_j by* (5.2)*. Then*

$$L_{\beta_2,n} \le L_{\beta_1,n} \le f(\lambda) \le U_{\beta_1,n} \le U_{\beta_2,n}, \qquad \lambda > 0. \tag{6.1}$$

Hence the smallest admissible order gives the strongest bounds in this hierarchy, at each truncation and each positive coupling.

Proof. For any positive measure ν_1 with the prescribed order-β_1 moments through degree k, let ν_2 be the law of SB where $S \sim \nu_1$ and $B \sim \mathrm{Beta}(\beta_1, \beta_2 - \beta_1)$ independently. Its moments are

$$\int s^j \, \mathrm{d}\nu_2 = \frac{(\beta_1)_j}{(\beta_2)_j} \frac{j! w_j}{(\beta_1)_j} = \frac{j! w_j}{(\beta_2)_j}, \qquad j \le k.$$

The beta identity (5.5) preserves the integral value. Thus every value attainable in the order-β_1 moment class is also attainable in the order-β_2 moment class. Taking infima for $k = 2n-1$ and suprema for $k = 2n$, and applying Lemma 6.1, proves (6.1). □

This is a uniform selection rule, with no estimated crossover parameter. It says that forgetting the beta factorization enlarges the possible values compatible with the coefficient data. At order one the rules are the familiar Stieltjes Padé bounds $[n-1/n]$ and $[n/n]$ [1]. At nonintegral orders they are positive sums of fractional powers, so they need not be rational functions of λ.

Theorem 6.3. *For the quartic and sextic cases of Theorem 5.2, the minimal-order bounds converge monotonically to $f_Q(\lambda)$ at every $\lambda > 0$. The same holds at every larger admissible order.*

Proof. The random variable $Q(\Omega)$ is bounded. Formula (5.3) therefore gives, for a fixed constant C, moments bounded by

$$\mu_{a,n} \le C^n \prod_{j=1}^{m-1} (a + j/m)_n.$$

For $m = 2, 3$, Stirling's formula implies $\mu_{a,n}^{-1/(2n)} \ge c\, n^{-(m-1)/2}$ for all sufficiently large n. The Stieltjes Carleman series diverges, so the representing measure is determinate. At higher order the beta factor is bounded by one and does not worsen this estimate.

By the extremal description, increasing n increases the lower bound and decreases the upper bound. The discrete Gauss and Radau measures are tight since their first moments are fixed. For any fixed moment, a higher fixed even moment bounds the tails uniformly as n grows; thus every weak limit has all the prescribed moments. Determinacy forces every such limit to be ρ_β. Finally, h_β is bounded and continuous, so both quadratures converge to its integral. □

7 A sextic consequence and a limit of the method

Let X be a standard real Gaussian and define

$$P_{m,k}(\lambda) = \frac{\mathbb{E}[X^{2k} e^{-\lambda X^{2m}/(2m)}]}{(2k-1)!!}, \qquad k \ge 0,$$

with $(-1)!! = 1$. This normalization gives $P_{m,k}(0) = 1$. Under the X^{2k}-weighted Gaussian measure, $X^2/2$ has gamma shape $k + 1/2$. The proof of Theorem 5.2 applies unchanged. Its exact order and representing variable are

$$a_k = \frac{2k+1}{2m}, \qquad T_{m,k} = (2m)^{m-1} \prod_{j=1}^{m-1} G_{a_k + j/m}. \tag{7.1}$$

All its moments come explicitly from the Wick coefficients

$$w_{m,k,n} = \frac{(2k + 2mn - 1)!!}{(2k-1)!!(2m)^n n!}. \tag{7.2}$$

Corollary 7.1. *For the scalar sextic model ($m = 3$), every normalized even correlation function admits convergent, certified bounds at all positive couplings:*

$$(2k-1)!! \frac{L^{(k)}_{a_k,n}}{U^{(0)}_{a_0,n}} \;\le\; \langle x^{2k} \rangle_\lambda \;\le\; (2k-1)!! \frac{U^{(k)}_{a_k,n}}{L^{(0)}_{a_0,n}}. \tag{7.3}$$

The numerator and denominator rules use (7.2)*. Their endpoints converge to the correlation function.*

Proof. Each factor is positive, and the same two-gamma Carleman argument in Theorem 6.3 applies for every fixed k. Divide the respective bounds and use $\langle x^{2k} \rangle = (2k-1)!! P_{3,k}/P_{3,0}$. □

The corollary supplies a concrete answer to part of the convergence question suggested by the even-potential direction in [7]. Its approximants are generally algebraic functions, rather than the rational Schwinger–Dyson approximants studied there. Convergence of those particular sextic truncations, and their possible product structure, remain separate questions.

There is also a precise reason not to extrapolate this argument to every degree. For $m \ge 4$, already the scalar minimal-order measure in (7.1) is a product of at least three gamma variables. It is moment indeterminate by the product criterion of Lin–Stoyanov [14, Corollary 1]. Thus distinct positive measures share all the reweighted Wick moments. The generalized Stieltjes transform is injective on finite measures: substitute $\lambda = 1/x$, divide the result by x^a, and write $(x + s)^{-a}$ as a gamma Laplace integral. Uniqueness of the two successive Laplace transforms proves the assertion. These measures therefore give different functions for some positive couplings, despite identical formal expansions.

This obstruction concerns reconstruction from that positive moment class. It does not make the original integral ambiguous. The differential equation, its analytic conditions, or the full gamma-product law (7.1) can supply information absent from the integer moments. In the proposed torsor approach, retaining such information is essential: not every useful restriction is equally informative, even when all its formal coefficients are known.

Exact relations also strengthen finite bounds in the determinate case. For the sextic model, integration by parts gives $\langle x^2 \rangle + \lambda \langle x^6 \rangle = 1$. An interval $[l_6, u_6]$ for the sixth moment therefore restricts the second moment to $[1 - \lambda u_6, 1 - \lambda l_6]$. Intersecting this with its independently computed interval retains information common to the moments. At $\lambda = 10$, the two-node rules give $0.0742 < \langle x^2 \rangle < 1.4575$; the intersection improves the upper bound to 0.9154. The archived calculation certifies both intervals. This elementary improvement uses an exact equation in addition to the separate positive moment representations.

8 Fields on rewrite graphs and hypergraphs

A rewrite system gives a directed graph whose vertices are states and whose arrows are individual rule applications. Equal states are merged; this convention matters when independent updates can be performed in either order. Such state graphs, including those of hypergraph rewriting, are studied in [23, Sections 5.3 and 6.2]. We use finite connected portions and their underlying undirected graphs. The field model below is an additional choice; the rewrite rules alone do not prescribe it.

Let H be a connected graph with $n \geq 2$ vertices and positive algebraic edge conductances c_e. Fix a vertex r, put $x_r = 0$, and set $d = n - 1$ and

$$E_H(x) = \sum_{uv \in E(H)} c_{uv}(x_u - x_v)^2 = x^{\mathsf{T}} L_r x, \qquad I_H(\lambda) = \int_{\mathbb{R}^d} e^{-E_H(x)/2 - \lambda E_H(x)^2/4} \, \mathrm{d}x.$$

Here L_r is the grounded weighted Laplacian. Write $\tau(H)$ for the weighted spanning-tree sum, $R_H(u, v)$ for effective resistance, and $\langle \cdot \rangle_{H,\lambda}$ for normalized expectation in this integral. The quadratic model at $\lambda = 0$ is a pinned Gaussian field on a network [16].

Proposition 8.1. *Define $Z_d(\lambda) = \mathbb{E} \exp(-\lambda |Y|^4/4)$ for a standard Gaussian $Y \in \mathbb{R}^d$. Then*

$$I_H(\lambda) = \frac{(2\pi)^{d/2}}{\sqrt{\tau(H)}} Z_d(\lambda), \tag{8.1}$$

$$\langle (x_u - x_v)^2 \rangle_{H,\lambda} = \frac{Z_{d+2}(\lambda)}{Z_d(\lambda)} R_H(u, v). \tag{8.2}$$

The exact positive-kernel order of Z_d is $a = d/4$, with

$$Z_d(\lambda) = \mathbb{E}(1 + 4\lambda G_{a+1/2})^{-a}, \qquad w_{d,j} = \frac{4^j (a)_j (a + 1/2)_j}{j!}. \tag{8.3}$$

If n is even, then

$$Z_d(\lambda) = \frac{q_{d-1}(\lambda)}{(d-2)!!}. \tag{8.4}$$

Consequently Z_d and the normalized even polynomial correlations are rational expressions in q_0, q_2. The same differential period torsor applies at every even vertex number.

Proof. The matrix-tree theorem gives $\det L_r = \tau(H)$; set $y = L_r^{1/2} x$. Rotational symmetry gives $\langle y y^{\mathsf{T}} \rangle = (Z_{d+2}/Z_d) I$. The usual network identity $(e_u - e_v)^{\mathsf{T}} L_r^{-1} (e_u - e_v) = R_H(u, v)$, with the rooted coordinate omitted, proves (8.2). Theorem 5.2 applied to $Q(y) = |y|^4/4$ proves (8.3), and radial integration proves (8.4). For odd d, recurrence (2.1) reduces every required even moment to q_0, q_2. For higher correlations the spherical moment formula gives Z_{d+2k}/Z_d times the Gaussian Wick sum of products of entries of L_r^{-1}. Finally, the adjacent moment pairs are related by

$$\begin{pmatrix} q_{2j+2} \\ q_{2j+4} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ (2j+1)/\lambda & -1/\lambda \end{pmatrix} \begin{pmatrix} q_{2j} \\ q_{2j+2} \end{pmatrix}.$$

The determinant is nonzero, so these pairs are rational changes of basis of the original rank-two connection. □

The graph factors have their own combinatorial descriptions. The resistance is the spanning two-forest sum separating u, v, divided by $\tau(H)$ [2, Theorem 2]. A killed-random-walk expansion of L_r^{-1} gives another convergent series [16, Chapter 2]. Either can be combined with the period bounds. If $L_d \leq Z_d \leq U_d$ and $L_{d+2} \leq Z_{d+2} \leq U_{d+2}$ are positive enclosures, then

$$R_H(u, v) \frac{L_{d+2}}{U_d} \leq \langle (x_u - x_v)^2 \rangle_{H,\lambda} \leq R_H(u, v) \frac{U_{d+2}}{L_d}.$$

Independent and dependent rewrites

Consider the string rule $A \to AB$. Starting from A^k, each state has the unique form $AB^{m_1} \cdots AB^{m_k}$, and one update increases one m_i. Retaining the states between A^k and $(AB)^k$ gives the oriented cube Q_k. It has 2^k states and $k!$ complete histories. Equivalently, represent a word by a directed path with edge colors A, B, and subdivide an A-edge into consecutive A- and B-edges. Isomorphism of these directed colored paths is equality of words. The examples are therefore also graph-rewrite systems with unnamed vertices. For $k = 2$ the two paths are

$$AA \longrightarrow ABA \longrightarrow ABAB, \qquad AA \longrightarrow AAB \longrightarrow ABAB.$$

The underlying graph is C_4. In contrast, $A \to AB \to ABB \to ABBB$ gives P_4, a four-state chain. Both use $Z_3 = q_2$, but their spanning-tree counts are 4 and 1, and their initial-to-terminal resistances are 1 and 3. Thus $I_{C_4} = I_{P_4}/2$, whereas the chain's endpoint variance is three times the diamond's at every positive coupling. At $\lambda = 1$, 16-node bounds give respectively $0.3777 < \langle (x_u - x_v)^2 \rangle_{C_4,1} < 0.3809$ and $1.1333 < \langle (x_u - x_v)^2 \rangle_{P_4,1} < 1.1425$.

Three independent rewrites already change the admissible approximation class: Q_3 has $d = 7$ and exact order $7/4$, so no ordinary Stieltjes representation exists. There is a finite certificate. At ordinary order one the candidate moments are $\mu_j = w_{7,j}$, and

$$\det(\mu_{i+j})_{0 \le i,j \le 2} = -\frac{3745374633}{65536} < 0.$$

At order $7/4$ all the required Hankel matrices are positive. At coupling one, the 16-node interval has width below 0.000841; order $11/4$, using the same coefficients, has width above 0.00437. The example gives an exclusion test and an optimal choice among the admissible orders.

For completeness, the cube's Laplacian eigenvalues are $2j$ with multiplicity $\binom{k}{j}$, as follows by using the characters of $(\mathbb{Z}/2)^k$. The matrix-tree theorem and the same characters give

$$\tau(Q_k) = 2^{2^k - k - 1} \prod_{j=1}^{k} j^{\binom{k}{j}}, \qquad R_{Q_k}(u, \bar{u}) = \frac{2}{2^k} \sum_{\substack{1 \le j \le k \\ j \text{ odd}}} \frac{\binom{k}{j}}{j}.$$

These formulas provide examples of arbitrary size within the same torsor.

Hypergraphs and the choice of interaction

For a finite hypergraph $\mathcal{H}$, use its incidence graph $B_{\mathcal{H}}$: vertices and hyperedges form its two vertex classes, and incidence gives its edges. This construction retains the hyperedges as separate objects. For $V = \{0, 1, 2, 3\}$ and $E = \{\{0, 1, 2\}, \{1, 2, 3\}\}$, the unit-weight incidence graph has six vertices, $\tau = 4$, and $R(0, 3) = 3$. Its partition function uses $Z_5 = q_4/3 = (q_0 - q_2)/(3\lambda)$ and has exact order $5/4$.

The square E_H^2 is a global interaction and has a precise limitation: at fixed n, integrating out the field weights graphs by $\tau(H)^{-1/2}$ independently of λ. The nonlinear coupling changes field correlations but not these relative graph weights. An edge-local quartic instead gives

$$\widetilde{I}_H(\lambda) = \int e^{-\frac{1}{2}\sum_e y_e^2 - \frac{\lambda}{4}\sum_e y_e^4} \, \mathrm{d}x, \qquad y_e = \sqrt{c_e}(x_u - x_v).$$

On a tree the d edge differences are independent Gaussian coordinates, so $\widetilde{I}_H = (2\pi)^{d/2} \tau(H)^{-1/2} q_0^d$. This is a symmetric-power expression in the same periods, again with exact order $d/4$. On graphs with cycles, this factorization generally fails, although Theorems 5.2–6.3 still apply. The accompanying example notes give the walk series, leaf-history coefficients, and checks for the edge-local P_4 model.

H	$\tau(H)$	$R(u,v)$	a	lower $Z_d(1)$	upper $Z_d(1)$
Q_1	1	1	1/4	0.77198088	0.77212212
$Q_2 = C_4$	4	1	3/4	0.36099863	0.36158743
P_4	1	3	3/4	0.36099863	0.36158743
Q_3	384	5/6	7/4	0.04460770	0.04544794
Q_4	42467328	2/3	15/4	0.00019404	0.00031685
$B_{\mathcal{H}}$	4	3	5/4	0.13660314	0.13747832

Table 1: Rewrite and incidence graphs with unit conductances. The resistance is between initial and terminal states, between antipodes for Q_k, or between vertices $0, 3$ in the hypergraph example. Bounds use 16 Gauss nodes and 17 Radau nodes at minimal order; endpoints are rounded outward.

9 Certified computations and practical choices

The accompanying program constructs (4.2) with rational truncated polynomials to obtain $w_{M,n}$, then forms $\mu_{\beta,n}$ by (5.2). Exact rational Hankel systems give monic orthogonal polynomials. Arb root isolation and positive Christoffel weights enclose the quadrature nodes, weights and evaluations. The program checks orthogonality and moment matching exactly or with certified balls, as appropriate. No fitted integral values enter the quadratures.

M	n	lower bound	upper bound	width $<$
1	2	0.74828962	0.79361826	4.54×10^{-2}
1	4	0.76598846	0.77778969	1.19×10^{-2}
1	16	0.77198088	0.77212212	1.42×10^{-4}
2	2	0.37044526	0.45678743	8.64×10^{-2}
2	4	0.39920223	0.41762905	1.85×10^{-2}
2	16	0.40725400	0.40732727	7.33×10^{-5}
3	2	0.11288594	0.20355949	9.07×10^{-2}
3	4	0.13291925	0.15093682	1.81×10^{-2}
3	16	0.13889499	0.13893524	4.03×10^{-5}
4	2	0.02091036	0.08766673	6.68×10^{-2}
4	4	0.02819444	0.04032442	1.22×10^{-2}
4	16	0.03080806	0.03082700	1.90×10^{-5}

Table 2: Minimal-order bounds for $Z_M(1)$. The Gauss rule has n nodes, and Radau has $n + 1$; moments through degree $2n$ suffice for the pair. Displayed endpoints are rounded outward. Widths bound the unrounded certified intervals.

The archived checks cover $1 \le M \le 4$, $1 \le n \le 16$, $\lambda = 1/100, 1/10, 1, 10, 100$, and orders $a, a + 1, a + 2$, together with ordinary order one when it is admissible. They verify the signs, monotonicity in n, and the order dominance in Theorem 6.2. Independent Bessel evaluations, inserted in (4.2), lie in every enclosure. The same program checks the two independent formulas for sextic representing moments and constructs bounds for the partition function and the second, fourth and sixth correlations. Independently enclosed convergent strong expansions lie in these bounds. These finite tests supplement the proofs; they are not the convergence argument.

Positive moment bounds are not always the fastest evaluation. For the quartic model we can

compare certified Taylor bounds directly. At size two let $w_n = w_{2,n}$ and define

$$B_n = \sum_{j=0}^{n} \frac{\Gamma(j/2+1/4)\Gamma((n-j)/2+3/4)}{j!(n-j)!}.$$

The two expansions and their degree-K absolute error bounds are

$$Z_2(\lambda) \sim \sum_{n\geq 0}(-\lambda)^n w_n, \qquad \epsilon_{\mathrm{G}} \leq \lambda^{K+1} w_{K+1},$$

$$Z_2(\lambda) = \frac{1}{\pi\lambda}\sum_{n\geq 0}(-\sqrt{2/\lambda})^n B_n, \qquad \epsilon_{\mathrm{Q}} \leq \frac{(2/\lambda)^{(K+1)/2} B_{K+1}}{\pi\lambda}.$$

The second bound follows by expanding the quadratic part under the positive product quartic measure, rather than by assuming that alternating terms decrease. At an ordinary center $\lambda = 1$, Cauchy's estimate on $|\lambda - 1| = R < 1$ gives

$$\epsilon_{\mathrm{T}} \leq \frac{(|\lambda-1|/R)^{K+1}}{1-|\lambda-1|/R}, \qquad |\lambda-1| < R,$$

because $|Z_M(\lambda)| \leq 1$ for $\Re\lambda \geq 0$. The Taylor coefficients use the seed periods at the center.

λ	Gaussian terms	quartic terms	Taylor terms at 1
0.01	10	591	—
0.1	—	91	—
0.9	—	29	11
1	—	27	1
1.1	—	26	11
10	—	13	—
100	—	8	—

Table 3: Terms sufficient to certify absolute error below 10^{-10} for Z_2, including Arb roundoff. Gaussian sums were searched through degree 128, quartic sums through degree 1024. A dash denotes failure of that bound in the search or exclusion from the chosen Taylor circle $R = 0.9$. Seed computation is excluded from the Taylor counts.

For an actual choice, first fix the available coefficient data and the cost to obtain new coefficients or seed periods. Within positive moment methods use order $M^2/4$. Then compare its certified interval with the weak, strong and ordinary Taylor bounds at the requested precision. The narrowest interval at a fixed budget is a reproducible criterion; it does not imply minimum running time without including coefficient, root-isolation and seed costs. At $\lambda = 1$, for instance, a short convergent quartic expansion is much more accurate than a 16-node moment rule. Moment rules instead give a seed-free method from Wick coefficients which extends beyond the rank-two special-function example. Rigorous special-function software already changes between expansions and continuation methods [10]. Constructive and generalized Borel methods provide other reorganizations of perturbation theory [18, 17]. Our comparison theorem selects the strongest positive moment bounds from the same coefficient data.

10 Arithmetic restrictions

The connection has an integral model over $\mathbb{Z}[1/2, t, t^{-1}]$. It can therefore be reduced at any odd prime. This is a change of base ring; a homomorphism $\mathbb{Q} \to \mathbb{F}_p$ does not exist. Such a reduction need not admit the characteristic-zero local frames.

Proposition 10.1. *With $\nabla = D - A$, the trace of the reduced p-curvature is*

$$\operatorname{Tr} \Psi_p = \frac{1}{4t^{2p}} \neq 0 \qquad (p \text{ odd}).$$

Consequently the reduced connection has no full horizontal frame over $\mathbb{F}_p((t))$ or $\mathbb{F}_p((t^{-1/4}))$ with their standard derivations.

Proof. For the determinant connection write $a = \operatorname{Tr} A = -t^{-1} - (4t^2)^{-1}$. Its p-curvature is $-(D^{p-1}a + a^p)$. Wilson's identity gives $D^{p-1}(-t^{-1}) = t^{-p}$; the double-pole term differentiates to zero. The simple-pole terms cancel with those of a^p, leaving the displayed value. Determinants commute with the connection, so this is the trace in rank two. In either displayed Laurent field $D^p = 0$. A full horizontal frame would trivialize the connection and force its p-curvature to vanish, a contradiction. □

Finite rational jets can still be reduced whenever their denominators are units. The companion computation verifies strong-basis jets through degree 24 at $p = 53, 59, 61$, and independently calculates the curvature at $p = 3, 5, 7, 11$. Prime-field data can reconstruct finite rational coefficients with denominator and height bounds; they do not directly approximate a real exponential period. A Frobenius or exponential-sum realization needs its own comparison theorem.

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